

Manoj Sai Penugonda

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SUMMARY

Data Scientist with 3.7 years of experience building scalable Machine Learning and Generative AI solutions. Strong expertise in data processing, feature engineering, predictive modeling, applied GenAI use cases, prompt engineering, transforming raw data into actionable insights that improve business outcomes and collaborating effectively with stakeholders for strategic decision-making.

EDUCATION

Anna University
Madras Institute of Technology

08/2018 - 07/2022
B.E. in Electronics Engineering

EXPERIENCE

Data Scientist
Accenture

07/2022 - Present
Chennai

Data Profiling

- Conducted **data profiling** on multiple datasets, resolving **1,500+** data quality issues and discrepancies- improving **data accuracy** by **25%** and reducing cleaning time by **20%**.
- Analyzed and documented **key data patterns**, enabling better data organization and improving data usability by **30%**.

Machine Learning & Predictive Modeling

- Built **regression-based ML models** that achieved **85% accuracy** in predicting weekly product sales volume based on promotional data and historical trends.
- Transformed inventory management strategies by reducing overstock instances by **18%** and stockouts by **14%**, improving overall inventory turnover rate.
- Engineered an ML-driven synthetic data generation pipeline leveraging generative models to produce **20,000+** highly realistic records, preserving **95% statistical fidelity** to the original dataset's **correlations** and **distributions**.
- Spearheaded **classification-based ML models** achieving an **88% AUC-ROC** score to predict patient readmission risk by analyzing complex clinical records, enabling early intervention for high-risk patients.

Text Similarity Search - RAG Framework

- Developed a **Generative AI**-based chatbot utilizing a **Retrieval-Augmented Generation (RAG)** framework, integrating **Hugging Face embeddings (thenlper/gte-small)** and **ChromaDB** to improve enterprise search accuracy by **25%** over traditional keyword methods.
- Executed semantic similarity search on historical text data, comparing user inputs against past records and retrieving **Top-K** matches with **95%** precision.
- Optimized vector search latency by **20%** through advanced text chunking strategies and refined similarity search parameters, enhancing retrieval speed and relevance for large-scale datasets.
- Designed a user-friendly GUI using **Streamlit**, reducing decision-making time for non-technical stakeholders by **40%** and improving accessibility.
- Led a **GenAI proof-of-concept (PoC)** to convert user stories into test cases, increasing project pipeline efficiency by **30%**.

TECHNICAL SKILLS

Programming and Query Languages: Python, SQL

Machine Learning & Deep Learning: Linear & Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, AdaBoost, Neural Networks (ANN, CNN, RNN), Transformers, Generative AI, LLM Fine-Tuning (LoRA), GANs, Diffusion Models, NLP, Clustering, Optimization Techniques

Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow

Tools & Technologies: Databricks, Azure, Git, GitHub, RESTful APIs, ReactJS, FastAPI

Analytics: Statistical Analysis, Data Visualization, Data Wrangling, Feature Engineering, A/B Testing, Hypothesis Testing

CERTIFICATIONS

- GH-300 GitHub Copilot.
- Certificate Program in Data Science and Machine Learning — IIT Delhi.